Mechanics of Human Movement

Week 10: Assignment 10

Due date on 2026-09-30, 23:59 IST

The due date for submitting this assignment has passed.Your last recorded submission was on 2026-09-18, 23:11 IST.

1. In persons with "Trendelenburg gait" the following gait deviation is observed: 1 Point

○ Hip hiking
○ Vaulting
● Lateral trunk bending
○ Steppage

Yes, the answer is correct. Score: 1

Accepted answer is Lateral trunk bending

2. A person is fitted with a locked knee orthosis. Which compensatory gait pattern is the person likely to adopt? 1 Point

☐ Anterior trunk bending
☐ Steppage
☑ Hip-hiking
☑ Circumduction

Yes, the answer is correct. Score: 1

Accepted answer is Hip-hiking, Circumduction

3. The metabolic cost of walking for a person with temporary pathological gait is most likely to _________compared to his/her own normal walking. 1 Point

● Be greater
○ Be lesser
○ Be the same
○ Depend on the type of pathological gait

Yes, the answer is correct. Score: 1

Accepted answer is Be greater

4. A person has weak hamstring muscles. Which compensatory gait pattern is the person likely to adopt? 1 Point

○ Anterior trunk bending
○ Steppage
○ Hip hiking
● Posterior trunk bending

No, the answer is incorrect. Score: 0

Accepted answer is Hip hiking

5. When do we see a maxima in the ankle joint moment during a normal gait cycle? 1 Point

○ Loading Response
○ Mid Stance
○ Mid Swing
● Terminal Stance

Yes, the answer is correct. Score: 1

Accepted answer is Terminal Stance

6. __________ is a stance phase gait modification. 1 Point

○ Circumduction
○ Steppage
○ Hip hiking
● Vaulting

Yes, the answer is correct. Score: 1

Accepted answer is Vaulting

7. Waddling gait pattern is observed when lateral trunk bending happens in_______ 1 Point

○ Right stance leg
○ Light stance leg
● Both the stance legs
○ Right leg swing phase

Yes, the answer is correct. Score: 1

Accepted answer is Both the stance legs

8. Which of the following are causes of pathological gait in humans? 1 Point

☑ Spasticity in muscles
☑ Muscle weakness
☐ Mild Fever
☑ Joint deformity

Yes, the answer is correct. Score: 1

Accepted answer is Spasticity in muscles, Muscle weakness, Joint deformity

9. Which of the following muscles are active during terminal swing phase? 1 Point

☑ Hamstrings
☐ Triceps Surae
☑ Tibialis Anterior
☑ Quadriceps

Yes, the answer is correct. Score: 1

Accepted answer is Hamstrings, Tibialis Anterior, Quadriceps

10. Functional leg length discrepancy 1 Point

☐ Is caused only by anatomical leg length discrepancy
☑ May be due to anatomical leg length discrepancy
☑ May be due to reasons other than anatomical leg length discrepancy
☐ Is not connected to anatomical leg length discrepancy

Yes, the answer is correct. Score: 1

Accepted answer is May be due to anatomical leg length discrepancy, May be due to reasons other than anatomical leg length discrepancy

11. Impaired proprioception is due to 1 Point

● Sensory issues
○ Muscular atrophy
○ Muscle spasticity
○ Pain

Yes, the answer is correct. Score: 1

Accepted answer is Sensory issues

12. A person walks with anterior trunk bending. What could be the potential causes for such a gait adaptation? 1 Point

☑ Weak knee extensors
☐ Weak hip extensors
☐ Weak ankle plantarflexors
☑ Hip flexion contracture

Partially Correct. Score: 0.67

Accepted answer is Weak knee extensors, Weak hip extensors, Hip flexion contracture

13. Steppage is a gait compensation adopted in case of inadequate 1 Point

● Dorsiflexion
○ Plantarflexion
○ Toe out
○ Toe in

Yes, the answer is correct. Score: 1

Accepted answer is Dorsiflexion